

- **Report Exercises: Logistics**



Applied Geodata Science II

Session 5

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14.10.2025



Course Plan

Course Website: https://geco-bern.github.io/agds2_course/

The screenshot shows a web browser window with the URL https://geco-bern.github.io/agds2_course/plan.html. The website has a dark header with the GECO logo and navigation links: Home, GitHub, a share icon, a lock icon, and a search icon. On the left, a sidebar lists 'About this course' with links to '1 Logistics', '2 Course plan' (selected), and 'References'. The main content area is titled '2 Course plan' and 'For fall semester 2025.' It features a section for 'Block 1: Introduction to Applied Geodata Science methods' with a paragraph explaining its purpose. Below this is 'Session 1 - 16.09.' with a note about the absence of lecturers and a link to the 'Applied Geodata Science book'. A bulleted list follows, mentioning 'Primers' and 'Reusable workflows'. On the right, a 'Table of contents' lists all course blocks and sessions.

Work

https://geco-bern.github.io/agds2_course/plan.html

GECO

Home

About this course

- 1 Logistics
- 2 Course plan
- References

2 Course plan

For fall semester 2025.

Block 1: Introduction to Applied Geodata Science methods

This first block serves to recap essential machine learning methods that will be applied in the tutorials of subsequent blocks, and to recap methods for implementing a reproducible and reusable data science workflow – required for the submission of the self-guided Report Exercises.

Session 1 - 16.09.

Note: No presence of lecturers. Self-study the following chapters in [Applied Geodata Science book](#):

- [Primers](#): Introduces basics of the R programming language and provides instructions for setting up the working environment with RStudio, including the installation of libraries used in this book.
- [Reusable workflows](#): Motivates the adoption of reusable, project-

Table of contents

- Block 1: Introduction to Applied Geodata Science methods
 - Session 1 - 16.09.
 - Session 2 - 23.09.
 - Session 3 - 30.09.
 - Session 4 - 07.10.
- Block 2: Phenology modelling
- Block 3: Digital soil mapping
- Block 4: Land cover classification
- Block 5: Spatial upscaling
- Block 6: TBA

Course Plan

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I - AGDS methods revision	1 Primers/Workflow/Code 16.09.25	9 Land cover classification 11.11.25 - Handful of Pixels Ch.7/ Ex.8.3	IV - Land cover
	2 Intro/Applications 23.09.25	10 Land cover classification 18.11.25 - Handful of Pixels Ch.7/ Ex.8.3	
	3 Supervised ML I + II 30.09.25	11 Spatial Upscaling 25.11.25 - Spat.Upsc. Ex.2	V - Upscale
	4 Random Forest + Interpr. ML 07.10.25	12 Spatial Upscaling 02.11.25 - Spat.Upsc. Ex.2	
II - Phenology	5 Phenology Modelling 14.10.25 - Handful of Pixels Ch.6/ Ex.8.2	13 TBA 09.12.25	VI - TBA
	6 Phenology Modelling 21.10.25 - Handful of Pixels Ch.6/ Ex.8.2	14 TBA 16.12.25	
III - Soil map	7 Digital soil mapping 28.10.25 - Tutorial DSM Ex.5		
	8 Digital soil mapping 04.11.25 - Tutorial DSM Ex.5		

FINAL REPORTS

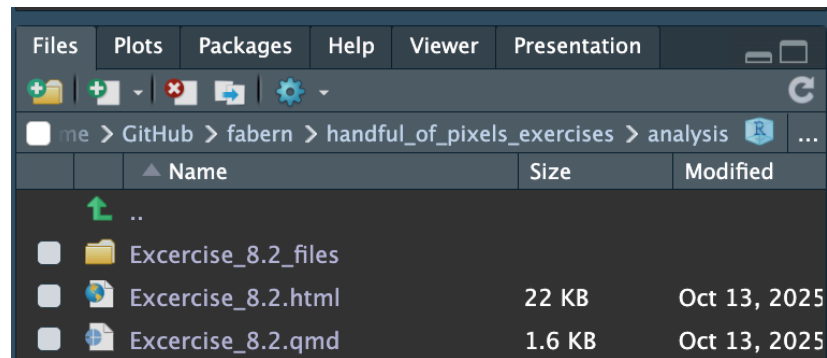
- Hand in as reproducible code (*git* repository)
- 2 out of 4 report exercises (separate reports as RMarkdown files)
- Including knitted HTML files
- Deadline for sending us the link by email: 23.12.2025

Logistics and Evaluation

Course Website: https://geco-bern.github.io/agds2_course/

CRITERIA

- **Reproducibility (20%).**
Make sure your RMarkdown code is reproducible and generates a html document as intended. You can check this by letting a colleague clone your git repository and run your code on their machine. If it fails, it's not reproducible, and you won't get points for this aspect.
- **Workspace organisation and code (20%).**
Follow the principles for workspace organisation and coding as described in the AGDS book.
- **Presentation (20%).**
This considers the appearance of the generated html document, its text and figures. Provide context and create good data visualisations.
- **Analysis and modelling (20%).**
This considers the correctness of analysis and modelling implementations, and the reasonable application of methods.
- **Interpretation (20%).**
This considers the correctness of the interpretation of results and answering to explicitly stated questions in the exercises.



FINAL REPORTS

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2 out of 4 report exercises - links on course website

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

7 Digital soil mapping

28.10.25 - Tutorial DSM Ex.5

11 Spatial Upscaling

25.11.25 - Spat.Upsc. Ex.2

9 Land cover classification

11.11.25 - Handful of Pixels Ch.7/ Ex.8.3

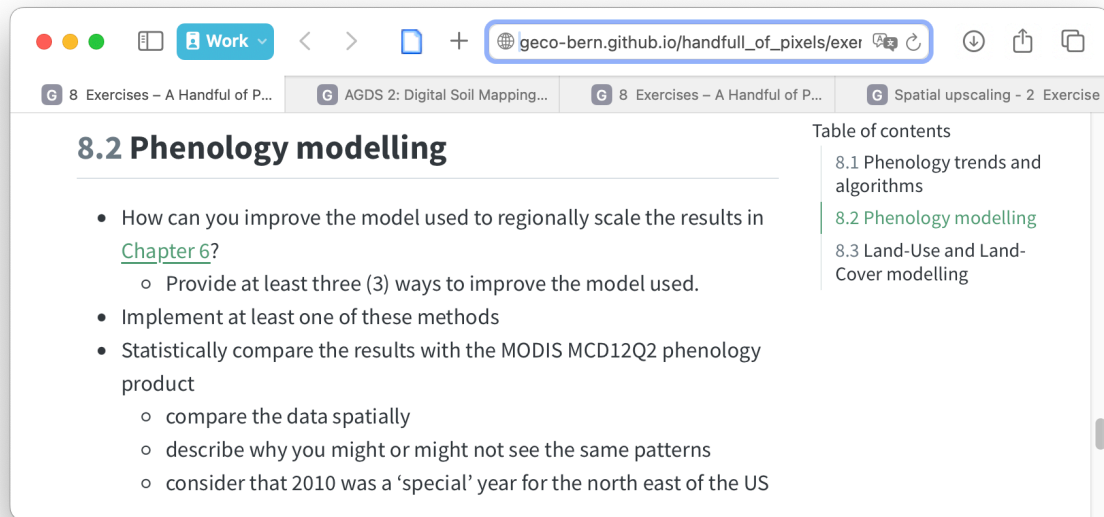
The collage shows several browser windows from the GECO course website. The visible windows include:

- 8.2 Phenology modelling**: A report exercise page with a table of contents and a list of tasks. The tasks include improving the model used to regionally scale results in Chapter 6, implementing at least three (3) ways to improve the model used, and statistically comparing the results with the MODIS MCD12Q2 phenology product. It also mentions that training data can be downloaded here and that submissions should be made as a pull request to the AGDS2 course repository.
- 5 Report Exercise**: A report exercise page for digital soil mapping. It includes a table of contents with sections 5.1 Simple model, 5.2 Variable selection, 5.3 Model optimization, and 5.4 Probabilistic predictions. The main text describes the tutorial demonstrated digital soil mapping for a continuous variable - top soil pH, and mentions that the observational data from file `data-raw/soildata/berne_soil_sampling_locations.csv` contains also categorical variables. Variables `waterlog_*` provide information about whether the soil was waterlogged at different depths (30, 50, and 100 cm). It is either `TRUE` (encoded in the model as `1`) or `FALSE`.
- 2 Exercise**: A report exercise page for spatial upscaling. It includes a table of contents with sections 2.1 Literature, 2.2 Random cross-validation, 2.3 Spatial cross-validation, and 2.4 Environmental cross-validation. The main text asks the user to read the paper by Ludwig et al. (2023) and answer the following questions: Explain the difference between a random cross-validation and a spatial cross-validation, and In spatial upscaling, we model the target based on environmental covariates. This implies that we assume the training data to sufficiently represent the conditions on which the model will be applied for generating predictions. Prediction errors may increase with an increasing distance of the prediction location from the training

2 out of 4 report exercises - links on course website

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2



The screenshot shows a web browser window with the address bar displaying `geco-bern.github.io/handfull_of_pixels/exercises`. The page title is "8.2 Phenology modelling". The main content area lists three bullet points:

- How can you improve the model used to regionally scale the results in [Chapter 6](#)?
 - Provide at least three (3) ways to improve the model used.
- Implement at least one of these methods
- Statistically compare the results with the MODIS MCD12Q2 phenology product
 - compare the data spatially
 - describe why you might or might not see the same patterns
 - consider that 2010 was a 'special' year for the north east of the US

A "Table of contents" sidebar on the right lists the following sections:

- 8.1 Phenology trends and algorithms
- 8.2 Phenology modelling
- 8.3 Land-Use and Land-Cover modelling

You will need:

- NetCDF library
- User account for NASA Earthdata Portal to download data
- **git**:
 - careful when working in OneDrive
make available offline
 - no need to use **gpg** (RStudio options)
 - use **ssh** link instead of **http** link
(check: **git remote -verbose**)
 - usual git cycle:
git pull
git commit
git push